

MTH2021 Mathematical Methods 2

Self-study sheet 3 – solutions

1.

$$(-1)^{i/\pi} = \exp((i/\pi) \log(-1)) \quad (1)$$

$$= \exp((i/\pi)i(\pi + 2k\pi)) \quad (2)$$

$$= \exp(-(1 + 2k)) \quad , \quad k = 0, \pm 1, \pm 2, \dots \quad (3)$$

2.

$$i^{1/3} = \exp\left(\frac{1}{3} \log i\right) \quad (4)$$

$$= \exp\left(\frac{1}{3}(\log|i| + i \arg i)\right) \quad (5)$$

$$= \exp\left(\frac{1}{3}(\log 1 + i(\pi/2 + 2k\pi))\right) \quad (6)$$

$$= \exp(i(\pi/2 + 2k\pi)/3) \quad , \quad k = 0, \pm 1, \pm 2, \dots \quad (7)$$

Distinct solutions arise for any three consecutive values of k , e.g. $k = 0, 1, 2$. These solutions are the same as what we would have obtained by the procedure for n -th roots.

3.

Write

$$\sin(x + iy) = \sin x \cos(iy) + \cos x \sin(iy) \quad (8)$$

$$= \sin x \cosh y + i \cos x \sinh y \quad (9)$$

Then for $u = \sin x \cosh y$ we have $u_{xx} = -u$, $u_{yy} = u \Rightarrow u_{xx} + u_{yy} = 0$.

Similarly for $v = \cos x \sinh y$ we have $v_{xx} = -v$, $v_{yy} = v \Rightarrow v_{xx} + v_{yy} = 0$.

4.

(a)

$$\left| \frac{\frac{z^{n+1}}{(n+1)!}}{\frac{z^n}{n!}} \right| = \frac{|z|}{n+1} \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad (10)$$

for all z and hence, by the ratio test, the series is absolutely convergent for all z .

(b)(i) We can multiply absolutely convergent series by forming the so-called Cauchy product (a particular way of grouping terms):

$$\exp z \exp w = \left(\sum_{k=0}^{\infty} \frac{z^k}{k!} \right) \left(\sum_{l=0}^{\infty} \frac{w^l}{l!} \right) \quad (11)$$

$$= \sum_{k=0}^{\infty} \sum_{l=0}^k \frac{z^l}{l!} \frac{w^{k-l}}{(k-l)!} \quad (12)$$

$$= \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{l=0}^k \frac{k!}{l!(k-l)!} z^l w^{k-l} \quad (13)$$

$$= \sum_{k=0}^{\infty} \frac{1}{k!} (z+w)^k = \exp(z+w) \quad (14)$$

(b)(ii) It is easiest to work from the other end:

$$\cos z + i \sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!} + \sum_{k=0}^{\infty} i(-1)^k \frac{z^{2k+1}}{(2k+1)!} \quad (15)$$

$$= \sum_{k=0}^{\infty} i^{2k} \frac{z^{2k}}{(2k)!} + \sum_{k=0}^{\infty} i^{2k+1} \frac{z^{2k+1}}{(2k+1)!} \quad (16)$$

$$= \sum_{k=0}^{\infty} \frac{(iz)^k}{k!} = \exp(iz) \quad (17)$$

5.

First note that if a complex series $\sum_k z_k$ is absolutely convergent, and $z_k = x_k + iy_k$ with x_k, y_k real, then $\sum_k x_k, \sum_k y_k$ are also absolutely convergent. This follows from $|x_k| \leq |z_k|, |y_k| \leq |z_k|$ and the comparison test. So consider

$$A = \sum_{k=0}^{\infty} p^k \exp(ik\theta) \quad (18)$$

$$= \sum_{k=0}^{\infty} (pz)^k, \quad z = \exp(i\theta) \quad (19)$$

A converges absolutely since $|pz| < 1$. Hence so do $C = \operatorname{Re} A$ and $S = \operatorname{Im} A$. Finally

$$A = \frac{1}{1-pz} = \frac{1}{1-p \exp(i\theta)} \quad (20)$$

Taking real and imaginary parts:

$$C = \sum_{k=0}^{\infty} p^k \cos k\theta = \frac{1-p \cos \theta}{1-2p \cos \theta + p^2} \quad (21)$$

$$S = \sum_{k=0}^{\infty} p^k \sin k\theta = \frac{p \sin \theta}{1-2p \cos \theta + p^2} \quad (22)$$

6.

Along the real axis: $z = x$, $dz = dx$, and the integral is

$$I_1 = \int_{-1}^1 \frac{x}{1+x^2} dx = 0 \quad (23)$$

Along the circle: $z^* = \exp(-it)$, $|z| = 1$, $dz = i dt \exp(it)$, and the integral is

$$I_2 = \int_0^\pi \frac{\exp(-it)}{1+1} i dt \exp(it) = \frac{\pi i}{2} \quad (24)$$

The integral around the closed path is $I_1 + I_2 = \pi i/2$.